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Implementation of Edge AI for Web-Based Atopic Dermatitis Skin Disease Classification Using Streamlit

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Abstract

Atopic dermatitis (AD) is a chronic skin disease with symptoms such as itching, dryness, and redness that interfere with patients quality of life. Its prevalence continues to increase, especially among children. Early diagnosis is crucial for effective treatment, but traditional methods such as allergy tests and biopsies are time-consuming and prone to error. This study proposes an automated classification system based on deep learning integrated with Edge AI for skin image analysis. Edge AI enables local processing, reduces latency, and safeguards data privacy. The system is implemented in a Streamlit-based web application for easy use by medical professionals. Evaluation was conducted using accuracy, precision, recall, specificity, and F1-score. Training results demonstrated high performance: ResNet-34 (96%), MobileNetV3 (98.5%), and EfficientNet-B0 (98.3%), respectively. A system demo is available at: https://www.youtube.com/watch?v=hkbXEbA8Gq0.

Keywords—Atopic Dermatitis, CNN, Deep Learning, Streamlit, Edge AI

1. Introduction

Atopic Dermatitis (AD) is a chronic, relapsing inflammatory skin disease characterized by intense itching, dryness, and erythema, often accompanied by significant psychological distress [1]. The multifactorial nature of AD requires a multidisciplinary management approach, including medical treatment, optimal skincare routines, and continuous education for patients and their families to minimize physical and psychosocial impacts. Globally, AD affects nearly 20% of children and 10% of adults, with an increasing prevalence over recent years [2]. The pathogenesis of AD involves a complex interplay of genetic predisposition, environmental triggers, and immune system dysregulation, commonly associated with elevated serum Immunoglobulin E (IgE) levels [3]. Typically, symptoms emerge within the first six months of life, although only around 25% of cases persist into adulthood. In the United States alone, the annual cost of managing AD exceeds $5 billion [4], while in Indonesia, approximately 15% of children are affected [5].

Several contributing factors such as urbanization, changing lifestyles, and increased exposure to environmental pollutants have been linked to the rising incidence of AD. In Indonesia, standard treatment typically involves the application of moisturizers and

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topical therapy, with evidence-based practices gradually being adopted [6]. The importance of early diagnosis is paramount in preventing complications and reducing symptom severity [7]. Conventional diagnostic techniques, such as allergy tests and skin biopsies, are widely used but come with limitations including long processing times and diagnostic uncertainty [8]. Accurately identifying symptoms such as erythema, skin lesions, and dryness presents a significant challenge, particularly in primary care settings.

To overcome these challenges, this study applies deep learning technologies to analyze skin images and precisely detect AD features. Deep learning technologies offer a promising solution, enabling non-invasive skin image analysis that can detect patterns often missed by the human eye. The system provides fast, consistent, and objective results, thus enhancing clinical decision-making.

Moreover, the integration of Edge Artificial Intelligence (Edge AI) in healthcare has been applied to overcome issues of high latency and bandwidth usage typically found in conventional AI models. By processing data locally using NVIDIA Jetson Orin Nano, Edge AI enables faster diagnosis, reduces the risk of data breaches, and provides greater autonomy in data handling [9].

This study implements a deep learning-based classification system for Atopic Dermatitis, integrated with Edge AI technology. The models compared in this study are EfficientNet-B0, ResNet-34, and MobileNetV3. The system is tested for classifying images of both normal and affected skin conditions and is accessible via a user-friendly web application built using the Streamlit framework. This approach provides an accurate, efficient, and accessible diagnostic tool to support medical professionals in identifying and managing AD.

2. Proposed Method

This research includes a series of stages systematically designed to build a skin disease classification system for Atopic Dermatitis (AD) based on deep learning and Edge AI. The process begins with the collection of datasets, namely images of normal skin and skin with AD, which are obtained from the open dermatology data source DermNet. After the data is collected, the image resizing stage is carried out to fit the standard input architecture of the deep learning models used, such as ResNet-34, EfficientNet-B0, and MobileNetV3. To increase the variety and generalization ability of the model, data augmentation techniques such as image rotation and flipping were applied. Next, a pre-processing stage including image normalization and light noise removal was performed to ensure optimal data quality before training. Once the data was ready, a classification model was trained to recognize typical visual patterns of AD, such as redness, dryness and lesions on the skin. The model is then evaluated using several performance metrics, such as accuracy, precision, recall, specificity, and F1-score, to assess how well the model distinguishes between normal skin and skin with AD. As a final stage, the model is integrated in an Edge AI environment, which allows the inference process to be performed locally (without the need for constant internet connection), thus speeding up the system response and maintaining the privacy of user data. The system is also equipped with a web-based application interface using Streamlit, which allows users, including medical personnel or patients, to upload skin images and obtain classification results directly through a simple and easy-to-use web interface. This approach is expected to support early detection and management of AD more efficiently and accurately. Detailed processes can be referenced in Figure 1.

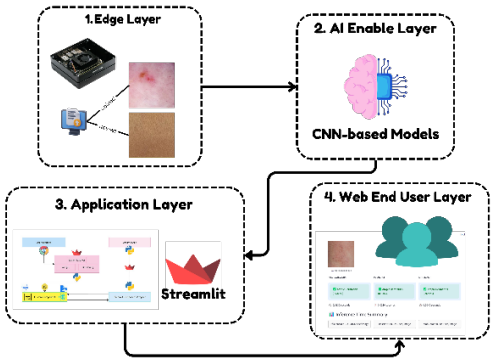

Figure 1: Proposed Flowchart

2.1. Data Collection

The materials used for research in this study include images of atopic dermatitis obtained from the DermNet dataset, as well as normal skin images derived from the USK-Normal Skin Dataset [10-11]. These images serve as the primary input for training and testing a deep learning-based classification model. The classification categories are divided into two classes, namely atopic dermatitis class and normal skin class. This binary classification approach is designed to detect visual differences between skin affected by atopic dermatitis and healthy skin, thereby supporting early detection.

Table 1: Dataset Specification

Dataset Name	USK-Normal Skin Dataset	DermNet
Total Dataset	170	500
Nation	Indonesia	Mixed
Resolution	600×450	640×480

The first step is image resizing. A standard size of 224 x 224 pixels, as recommended for all three architectures, is

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used. The next step is image data augmentation. This augmentation process is applied to the collected atopic dermatitis dataset. It is performed to increase the amount of training data and reduce the risk of overfitting. Augmentation is applied to 500 atopic dermatitis skin images. The process includes rotation of 10 and 90 degrees, zooming by 20%, shifting by 10 degrees, as well as horizontal and vertical flipping. The number of images before and after augmentation is presented in Table 2 below.

Table 2: Dataset after Augmentation

Dataset Name	Normal Skin	Atopic Dermatitis Skin	Total
Before Augmentation	170	500	670
After Augmentation	3000	3000	6000

2.2. Edge AI

Edge AI was used to optimize the image classification process, with experiments conducted on a Jetson Orin Nano (8GB RAM, 15W power mode) [12]. The integration of machine learning (ML) and deep learning (DL) algorithms into edge devices facilitates the processing of large volumes of health data, including electronic medical records and real-time monitoring data from wearable devices [13]. The full workflow is shown in Figure 1, while the hardware specifications are listed in Table 3.

Table 3: Overview of Jetson Orin Nano [14]

Name	Jetson Orin Nano 8GB
Released Time	January 2023
AI Performance	40 TOPs
GPU Detail	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores (Orin Nano 8GB)
CPU Detail	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3
Power	7-15 W
Memory	8GB 128-bit LPDDR5 68 GB/s (Orin Nano 8GB)

One of the main advantages of edge AI is its ability to reduce latency in data processing, which is crucial in medical emergencies where every second counts [15]. In addition, edge AI enhances data privacy and security by minimizing the need to transfer sensitive information to centralized servers, thereby reducing the risk of data breaches [16]. However, the implementation of edge AI in the healthcare sector faces several challenges, such as high resource requirements, interoperability issues, and the complexity of integrating various technologies. These challenges must be addressed to ensure the effective deployment of edge AI.

2.3. Convolutional Neural Network

Convolutional Neural Network (CNN) is one of the most popular deep learning algorithms [17]. CNN is an advancement of the Multi-Layer Perceptron (MLP), specifically designed to process and recognize data in the form of two-dimensional images. This algorithm is classified as a feedforward artificial neural network, where at least one of its layers utilizes convolutional operations inspired by the functioning of biological neural networks. Convolutional Neural Networks (CNNs) integrate artificial neural networks with discrete convolution to autonomously extract features from image data. In contrast to traditional image recognition algorithms, CNNs accept raw image data as input, bypassing the need for complex feature extraction and data reconstruction processes. The architecture of a CNN typically consists of four key layers: convolutional layers, pooling layers, activation functions, and fully connected layers. In this research, ResNet-34, EfficientNet-B0, and MobileNetV3 CNN architectures were used to differentiate between atopic dermatitis and normal skin images, utilizing both original datasets. Figure 2 illustrates the CNN architecture from input to final output.

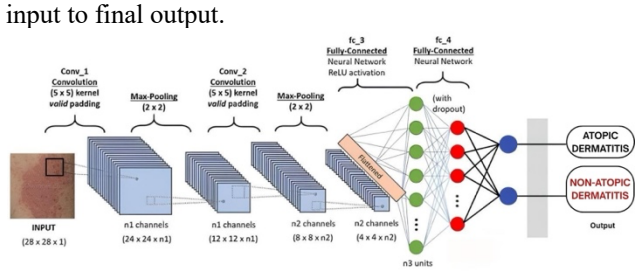

Figure 2: Convolutional Neural Network (CNN)

2.4. Streamlit Web-App

Streamlit is a Python library that has gained significant popularity. This framework is specifically designed by Python to simplify the process of creating dashboard interfaces, which are commonly used in Machine Learning and Data Science applications. Streamlit is highly compatible with various Python libraries such as Scikit-learn, Keras, PyTorch, SymPy (latex), NumPy, pandas, Matplotlib, and others. Its usage is straightforward, as it allows for direct HTML commands to be written within Python files. There is no need to use additional libraries for styling, as all visual settings are handled by Streamlit. This makes development with Streamlit faster and more efficient.

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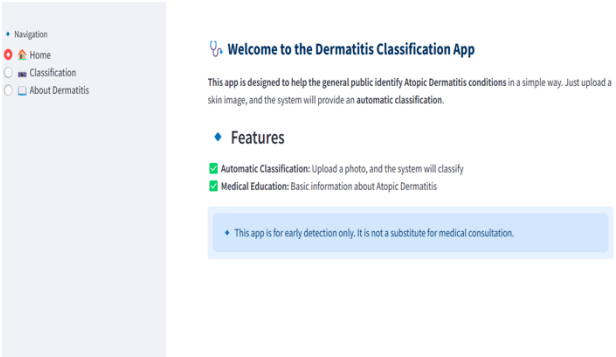

Figure 3: User Interface of Streamlit

2.5. Model Training

This research utilized ResNet-34, EfficientNet-B0, and MobileNetV3 architectures to design the model. These architectures were chosen for their ability to enhance CNN performance by optimizing network width, depth, and resolution, while keeping the number of parameters relatively minimal. The experiments were carried out using the PyTorch framework on the Google Colaboratory platform. The model's input images consisted of original RGB images of both atopic dermatitis and normal skin, with dimensions of 3×224×224 pixels.

Table 4: Hyperparameter of The Training Process

Parameter	EfficientNet -B0	ResNet-34	MobileNet V3
Epoch	50	50	50
Batch Size	2	2	2
Loss Function	Cross-Entropy	Cross-Entropy	Cross-Entropy
Learning Rate	0.01	0.01	0.01
Optimizer	SGD	SGD	SGD
Momentum	0.9	0.9	0.9

2.6. Evaluation Matrix

In classification problems involving two classes, each instance is assigned to either a positive or negative label. A classifier maps instances to predicted labels, which may be continuous or discrete depending on the model. To distinguish between the true label and the predicted label, the notations {p, n} for actual labels and {Y, N} for predicted outputs are used. Based on these outcomes, a confusion matrix can be built with four components: true positive (TP), false positive (FP), true negative (TN), and false negative (FN), providing a structured way to evaluate model performance [18].

3. Result and Discussion

The results of this study focus on the classification of atopic dermatitis and normal skin using deep learning models trained on original skin images. The classification process employed ResNet-34, EfficientNet-B0, and MobileNetV3 architectures.

3.1. Edge AI Deployment

The system was implemented on an edge application using the Streamlit web framework, as illustrated in Figure 4. The web application is capable of displaying classification results for both normal skin and atopic dermatitis. On the Jetson Orin side, the trained model from Google Colaboratory was converted into the ONNX (Open Neural Network Exchange) format. This conversion aims to optimize GPU memory usage and improve adaptability when handling various input batch sizes.

Figure 4: Physical Setting of Edge AI Web-Based Application

Additionally, as shown in Figure 5 and Figure 6, the Streamlit interface presents a user-friendly platform for real-time interaction, enabling users to easily upload and analyze skin images, with the results displayed promptly for medical evaluation.

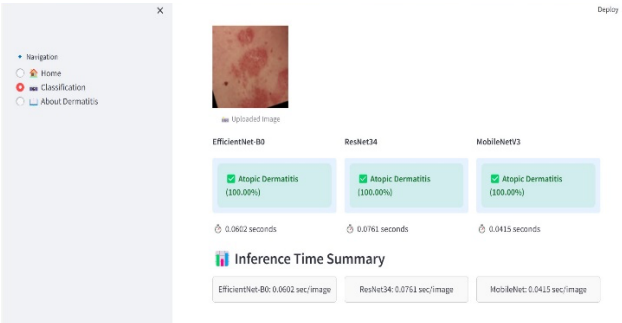

Figure 5: Main Interface of the Classify Atopic Dermatitis, Implemented on the Edge-AI Jetson Orin Nano

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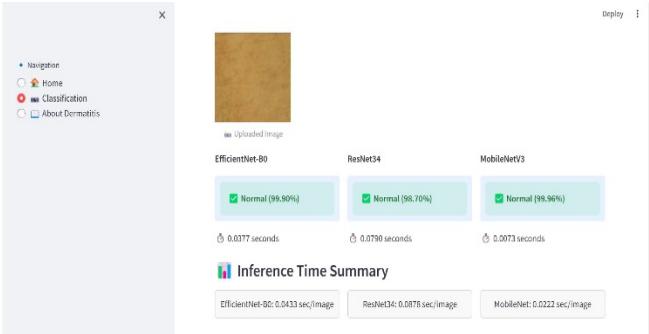

Figure 6: Main Interface of the Classify Normal Skin, Implemented on the Edge-AI Jetson Orin Nano

3.2. Model Perfomance

At this stage, analysis and evaluation are conducted to measure the performance of the designed, trained, and tested model, using evaluation metrics based on the confusion matrix.

Table 5: Evaluation Matrix Result

	ResNet-34	EffNet-B0	MobileNetV3
Training time	88m 25s	**50m 71s**	90m 20s
Acc	96%	98.3%	**98.5%**
Rec	94%	97.8%	**99%**
Spec	**99%**	98%	98.1%
F1-Score	96.6%	98.3%	**98.4%**

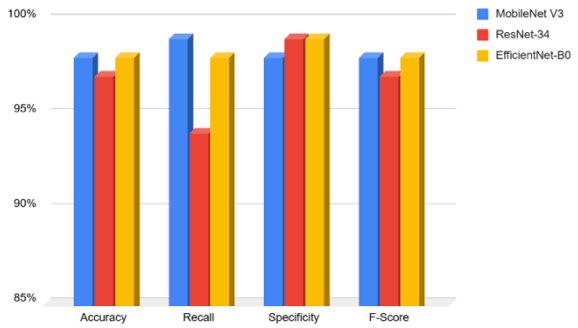

Figure 7: Comparison of Evaluation Metric Result

From Figure 7, it can be observed that all three CNN architectures ResNet-34, EfficientNet-B0, and MobileNetV3 demonstrate high performance in classifying images of normal and atopic dermatitis skin. This is evident through the evaluation metrics, including accuracy, recall, specificity, and F1-score. The ResNet-34 model achieved 97% accuracy, 94% recall, 99% specificity, and an F1-score of 96.6%. Meanwhile, EfficientNet-B0 outperformed ResNet-34 with 98.3% accuracy, 97.8% recall, 98% specificity, and an F1-score of 98.3%. MobileNetV3 recorded the best overall performance, reaching 98.5% accuracy, 99% recall, 98.1% specificity, and an F1-score of 98.4%. These results indicate that MobileNetV3 is the most effective architecture among the three, with the highest classification performance. Furthermore, although MobileNetV3 had the longest training time of 90 minutes and 20 seconds, ResNet-34, despite requiring 88 minutes and 25 seconds, still produced better results, demonstrating that longer training time does not necessarily equate to lower performance.

3.3. Quantitive Measurements

All models were trained using the same parameters, such as 50 epochs, to ensure fair in performance comparison and avoid model bias. The purpose of this standardization was to validate that the evaluation results accurately reflect the capability of each model architecture.

Figure 8: Comparison of Training Accuracy and Validation Accuracy

Figure 8 presents a bar chart showing training accuracy and validation accuracy for three deep learning models: EfficientNet-B0, ResNet-34, and MobileNetV3. For training accuracy, MobileNetV3 achieved the highest value of 0.985, followed by ResNet-34 at 0.96, and EfficientNet-B0 at 0.983. This indicates that all three models performed well during training, with MobileNetV3 slightly outperforming the others in terms of training accuracy. Regarding validation accuracy, MobileNetV3 recorded the highest value of 0.996, followed by EfficientNet-B0 at 0.99, and ResNet-34 at 0.9283. This suggests that MobileNetV3 has the best ability to generalize to unseen data, while ResNet-34 showed a lower value on the validation dataset.

MobileNetV3 exhibited the lowest training loss of 0.001, with ResNet-34 following at 0.007, and EfficientNet-B0 at 0.011. These results indicate that MobileNetV3 trained most efficiently among the models.

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In terms of validation loss, MobileNetV3 also demonstrated the best performance, with a value of 0.038, followed by EfficientNet-B0 at 0.07, and ResNet-34 at 0.2232. The relatively lower validation loss values for MobileNetV3 and EfficientNet-B0 suggest superior generalization, while ResNet-34 faced greater difficulty maintaining consistency during validation.

4. Conclusion

This research focuses on deploying deep learning models on edge AI devices for the classification of normal skin and atopic dermatitis. The models evaluated include ResNet-34, EfficientNet-B0, and MobileNet-V3. Their performance was assessed using several metrics, including accuracy, precision, recall, specificity, and F1-Score. Among the tested models, MobileNet-V3 achieved the highest accuracy of 98.5%, followed by EfficientNet-B0 with 98.3%, and ResNet-34 with 97%. These results show that all models performed effectively in distinguishing between normal skin and atopic dermatitis. Nevertheless, MobileNet-V3 demonstrated the most outstanding performance, establishing it as the most reliable model in this research.

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